Lemma. Let $X : \mathcal{U}(\subset \mathbb{R}^2) \to S$ be a parametrization of a regular surface S. Given $q \in \mathcal{U}$,

$$DX_q[\mathbb{R}^2] = \{\alpha'(0) \mid \alpha : (-\varepsilon, \varepsilon) \to S \text{ is differentiable and } \alpha(0) = X(q)\}.$$

Proof. Let w be a tangent vector at $X(q)$. That is, for some smooth curve $\alpha : (-\varepsilon, \varepsilon) \to X[\mathcal{U}] \subseteq S$ such that $\alpha(0) = X(q)$, $w = \alpha'(0)$. Since $X^{-1} : X[\mathcal{U}] \to \mathbb{R}^2$ is differentiable, the curve $\beta = X^{-1} \circ \alpha : (-\varepsilon, \varepsilon) \to \mathcal{U}$ is also differentiable, so $DX_q \cdot \beta'(0) = DX_q \cdot DX^{-1}_{X(q)} \cdot \alpha'(0) = \alpha'(0)$.

Conversely, let $w = DX_q(v)$ for some $v \in \mathbb{R}^2$. Then, a curve

$$\gamma : (-\varepsilon, \varepsilon) \to \mathcal{U} : t \mapsto tv + q$$

satisfies that $\gamma'(0) = v$ and $\gamma(0) = q$. Now, a curve $\alpha = X \circ \gamma$ satisfies $\alpha(0) = X(q)$ and $\alpha'(0) = DX_q \cdot v = w$, so proved ☒

Def. Let $S \subset \mathbb{R}^3$ be a regular surface and $p \in S$ be a point. Then, for a parametrization $X : \mathcal{U} \to S$ at $p \in S$, a unit vector

$$N(q) = \frac{X_u \times X_v}{|X_u \times X_v|}(q) \quad \text{where } q = X^{-1}(p)$$

is called an unit normal vector at p.

「Def. Given a regular surface S and a point $p \in S$, by definition, there is a parametrization $X : \mathcal{U} \to S$ containing p, ($p \in X[\mathcal{U}]$)
Then, for $q = X^{-1}(p) \in \mathcal{U}$, the Jacobian $dX_q : \mathbb{R}^2 \to \mathbb{R}^3$ is linear with rank 2.
The plain $dX_q[\mathbb{R}^2]$ is called a tangent plane at p for S, denote T_pS.

The tangent plane T_pS is independent of choice of the parametrizatin X.
But, given a parametrization X, T_pS is determined by a basis

$$\left\{ \frac{\partial X}{\partial u}(q), \frac{\partial X}{\partial v}(q) \right\},$$ it is denoted simply $\{X_u, X_v\}$.

Given a parametrization $X : \mathcal{U} \to S$ at $p \in S$, determine a vector $w \in T_pS$.
Given $w \in T_pS$, there is a curve $\alpha : (-\varepsilon, \varepsilon) \to X[\mathcal{U}]$ satisfying $\alpha(0) = p$, $\alpha'(0) = w$.
Let $\beta = X^{-1} \circ \alpha : (-\varepsilon, \varepsilon) \to \mathcal{U}$. It is also smooth, and $\alpha = X \circ \beta$. Now,

$$\alpha'(0) = \frac{d}{dt}(X \circ \beta)(0) = \frac{d}{dt} X(u(t), v(t))(0) = X_u(q) \cdot u'(0) + X_v(q) \cdot v'(0)$$

where $q = X^{-1}(p)$ and $\beta(t) = (u(t), v(t))$.

Let S be a regular surface and $V \subseteq \mathbb{R}^3$ be an open set such that $S \subseteq V$. If $f: V \to \mathbb{R}$ is differentiable, then for any point $p \in S$ and parametrization $X: \mathcal{U} \to S$ at $p \in S$, a map $f \circ X: \mathcal{U} \to \mathbb{R}$ is differentiable.

Height function.

Given a surface S and a fixed unit vector $v \in \mathbb{R}^3$, a function

$$h: S \to \mathbb{R};\ p \mapsto \langle p, v \rangle$$

is called a height function with respect to $v \in \mathbb{R}^3$.

「Unit Sphere.

Let $S^2=\{(x,y,z)\in\mathbb{R}^3 \mid x^2+y^2+z^2=1\}$ be given. Define a map

$$R_{z,\theta}:\mathbb{R}^3\to\mathbb{R}^3;(x,y,z)\mapsto(\cos\theta\cdot x-\sin\theta\cdot y,\ \sin\theta\cdot x+\cos\theta\cdot y,\ z).$$

(That is, z-rotation of degree θ). Since $\cos\theta$, $\sin\theta$ are just constant, $R_{z,\theta}$ is differentiable. So, the restriction $R_{z,\theta}|_{S^2}$ is also differentiable

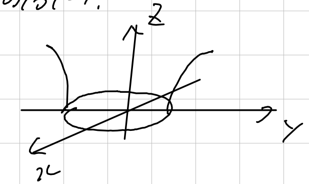
Calculate the tangent plane at a point $p\in S^2$. Simply, denote $R=R_{z,\theta}$. Since S^2 is regular surface, there is a smooth curve $\alpha:(-\varepsilon,\varepsilon)\to S^2$ such that $\alpha(0)=p$.

Problems, Do Carmo sec 2.4.

Explicit the tangent plane of $x^2+y^2-z^2=1$ at $(x_0, y_0, 0)$.
suppose that $x_0, y_0 > 0$. Consider a patch

$$X : D \to S ; (u,v) \mapsto \left(\sqrt{1-u^2+v^2},\, u, v\right)$$

where D is unit disk.



「Excersice,

Sec 2.4, 11.

Show that a normal line of a surface parametrized by

$$X(u,v)=(f(u)\cdot\cos v,\ f(u)\cdot\sin v,\ g(u)),\quad f(u)\neq 0,\ g'\neq 0.$$

must across the Z-axis.

Proof. Since the normal vector is determined by ; for each $q=(u,v)$,

$$N(q)=(X_u\times X_v)(q) \quad \text{and} \quad X_u=(f'(u)\cos v, f'(u)\sin v, g'(u))$$
$$X_v=(-f(u)\sin v, f(u)\cos v, 0).$$

So

$$N(q)=(-g'(u)f(u)\cos v, -g'(u)f(u)\sin v, f'(u)f(u)).$$

Now, the normal line is

$$L(t)=X(q)+t\cdot N(q)=(f(u)\cos v(1-t\cdot g'(u)), f(u)\sin v(1-tg'(u)), g(u)+tf'(u)f(u))$$

across the Z-Axis when $t=1/g'(u)$.